

**A Clinical Study on Islanders to Determine the Potential Effects
of Rest and Exercise on Memory**

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1 Abstract

Creating and recalling memories is an important function necessary for daily life. We aim to investigate lifestyle choices that can improve our abilities to encode, store, and retrieve memory. Research suggests sleep and exercise both have a significant impact on our working memory, also referred to as short-term memory. This is what we rely on to complete tasks where we need to hold and manipulate information temporarily. Using a randomized complete block design with a 2x2 factorial, we collected data on a sample of 200 Providence islanders and recorded their change in working memory performance with the influence of rest and running. Islanders are tasked with the Memory Game with their time recorded in seconds. They then complete their assigned rest task then their assigned run task with a Memory Game immediately following each task. The data is analyzed with a two-way ANOVA with age as a blocking factor. We found that there is a statistically significant main effect of rest, where more rest improved performance on the Memory Game. There was no significant main effect of exercise nor interaction between the two factors. Age, however, is a significant block. Our results on this study of effects on short-term memory can be extended to improving other forms of long-term memory.

2 Introduction

Memories are essential in creating an identity for oneself. Memory can be simply defined as the ability to recall information. Memories are responsible for the stories we tell ourselves and help build the basis of our relationships. Over time memory loss can lead to a feeling of losing yourself. Losing your skill to maintain a memory hurts basic yet essential daily tasks. Memory is essential for knowing your way around a neighborhood or remember to keep in contact with a distant cared for friend or family member.

Maintaining memories is a complex system. Sensory components of memory's such as sight, smell, and, etc. are distributed in different areas of the brain. To revoke these memories the brain is able to combine the different aspects to form the full memory. Every memory that is formed the constituent parents are stored in the hippocampus. Next to the hippocampus is the amygdala which is responsible for the emotions. Memories that can be considered important or emotionally significant are flagged by the amygdala. Differing components of a memory are then distributed to sections of the cerebral cortex, located in the outer layer of the brain.

Once these memories are stored the brain relies on the frontal lobes to retrieve the past memory. The frontal lobes rely on attention and focus. The multiple different aspects of a memory when combined together form a unique neuronal pattern that are dormant until recalled from memory, meaning it was reactivated. Given the significance memory has for an individual this study aims to measure factors such as sleep and exercise and the impact it has on memory. Academic research has found that both exercise and sleep help improve memory. It has been found that physical activity improves cognitive health because of the lack of loss of brain tissue during aging. Additionally, more physically active people have a higher capability of allocating attention and processing information. Similarly, studies have found that sleep can 're-organize' new memories.

After encoding information to store as memory, sleep is important to the consolidation of this memory to ensure its storage and retrieval (Kuriyama et al., 2008). A study conducted to test working memory performance found those in the sleep condition performed significantly better (Kuriyama et al., 2008). The researchers ensured that the results did not vary based on other factors that come with manipulating sleep, including the timing of sleep and circadian rhythm

effects as well as attention levels (Kuriyama et al., 2008). The results imply that there is a memory process that is enhanced during sleep and could play a role in expanding the capacity of working memory (short-term memory).

Exercise also enhances human memory with research supporting its effects on different forms of long-term memory: declarative memory, which includes information about facts and events; and procedural memory, which includes practiced motor skills (Roig et al., 2022). There are also short-term effects that have the most influence on cognitive performance within the first 10 to 20 minutes of exercising (Park & Schott, 2022). This is the duration we used for our experimental design to optimize the effects of exercise on working memory. The effect of exercise decays gradually over time after 20 minutes (Park & Schott, 2022).

To find an interaction effect of sleep and exercise on memory, Roig et al. measures participant performance on a recognition memory task, requiring participants to identify pictures presented after completing their assigned condition (2022). The four groups being only nap, only exercise, both nap and exercise, and neither nap nor exercise. The group with combined effects of exercise and nap showed greater recall abilities when identifying the pictures shown during the study phase than participants in the nap only and exercise only groups (Roig et al., 2022). The researchers also highlight that previous research also have similar results with memory tests after one or more nights of sleep show more influence of exercise on memory than after no sleep (Roig et al. 2022). Our study differs from research by Roig et al. in 2022 in the order we task the participants to either sleep or exercise. We chose to have participants rest before they exercise to find interaction effects of sleep on the exercise as well as memory instead of only memory performance itself. Also, with research by Park & Schott in 2022, we inferred that there would be enhanced effects on memory following exercise instead of delaying by a round of rest.

Both of these behaviors—physical activity and sufficient rest—are shown to enhance different stages of memory, including encoding, consolidation, and retrieval. However, few studies have explored the combined effect of short bouts of exercise and short-term rest, particularly in controlled experimental settings across different age groups. In this study, we aim to address that gap by analyzing the individual and combined effects of brief physical activity and short rest periods on memory performance. We utilize a randomized complete block design to measure how different rest conditions (napping vs. sitting) and exercise conditions (1km run vs. 100m run) influence memory, as measured by the time taken to complete a card-matching task. By blocking on age, we control for variability that might arise from cognitive differences due to aging. Ultimately, our goal is to determine whether rest and exercise, either independently or in combination, lead to measurable improvements in memory function—and whether such effects differ by age group.

3 Methods

3.1 Participants

Participants in this study will be islanders. We will assign each of the three islands a number and then use R to randomly select a number. For each data point, a city will be randomly selected using the same strategy. The seed will be set at 100 for both of these. When it comes to study participants, due to age-based blocking, random selection will be difficult. Instead, we will parse through households until an islander of the appropriate age range is found. We do not aim to exclude populations to have a representative sample for the whole island.

3.2 Design

The study will be a randomized complete block design with 2 x 2 factorial. The parameters for the design are as follows:

Response	Time (seconds)	Time to Complete Memory Game
Predictors	Run Type	Run Indoors 1km, Run Indoors 100m
	Rest Type	Nap 15 minutes, Sit 10 minutes
Block	Age	18–25, 26–35, 36–45, 46–55, 56–65

Figure 1: Parameters

The factor diagram is as follows:



Benchmark	Rest	Exercise	Interaction	Block (Age)	Error
1	1	1	1	4	192

Figure 2: Factor Diagram

We chose to focus on rest and exercise based on previous proposed impacts on boosting memory. We aim to determine if one of those actions involves memory growth. Finally, we will block with age since it is known that memory can get worse with age.

3.3 Instruments

To conduct our study, we assigned virtual tasks to virtual people on the Island. We selected our participants from the island to gather our sample. We measured the memory levels of the Islanders through the “Memory Game” option which entailed Islanders pairing 30 cards together as quickly as possible. The time it took for them to complete this task was recorded in seconds. Four different treatments were administered to the islanders and their impact on the time it takes to complete the Memory Game were recorded. These treatments are as follows: “Run Indoors 100m,” “Run Indoors 1km,” “Nap 15 mins,” and “Sit 10 mins.” We used Google Sheets to record data collected on the Island. To conduct our analysis and create our models, we used base R and select packages in R Studio.

3.4 Procedure

Step 1: Use a random number generator and multistage sampling method to (pseudo) randomly find subjects for the experiment.

- First Stage: Randomly select a town on Providence Island
- Second State: Randomly select a house within that town.
- Third Stage: Randomly select a person within that house

Repeat this until there are 200 subjects with 40 islanders in each of the following age groups: 18-25 year olds, 26 - 35 year olds, 36 - 45 year olds, 46 - 55 year olds, and 56 - 65 year olds.

Step 2: Use a random number generator to randomly assign half the islanders to each of the exercise conditions. Use the random number generator again again to assign half the islanders in each of the exercise conditions to each of the rest conditions. This should create four treatment groups of 10 islanders within each age block as follows:

1. Run indoors 1km, sit 10 mins
2. Run indoors 1 km, nap 15 mins
3. Run indoors 100 m, sit 10 mins
4. Run indoors 100 m, nap 10 mins

Step 3: For each islander, record the time it takes to complete the Memory Game.

Step 4: For each islander, apply the “Rest” treatment by having them either sit for 10 minutes or nap for 15 minutes.

Step 5: For each islander, record the time it takes to complete the Memory Game again.

Step 6: For each islander, apply the “Exercise” treatment by having them either run 1 kilometer indoors or run 100 meters indoors.

Step 7: For each islander, record the time it takes to complete the Memory Game.

Step 8: For each islander, compute the difference in time it takes to complete the Memory Game for after the exercise and before the rest, after the rest and before the rest, and after the exercise and after the rest. These will serve as response variables.

4 Data Analysis

4.1 Type of Statistical Analysis

To evaluate the effects of rest and exercise on memory performance, we used a two-way ANOVA with blocking. Our two main factors of interest are Rest Type (Nap vs. Sit) and Run Type (Run 1km vs. Run 100m). We are particularly interested in whether either of these factors, or their interaction, leads to a significant difference in the time it takes participants to complete the memory game. Blocking by Age Group (five levels from 18 to 65) allows us to account for expected differences in memory across age ranges. Including this blocking factor improves the precision of our comparisons by reducing variability within groups.

The ANOVA model tests for; the main effect of Rest Type, Run Type, the interaction between Rest and Run and the effect of Age Group as a block. Our baseline comparisons are made using the group that sat for 10 minutes and ran 100 meters, as this represents the least rest and the least exertion. This baseline allows us to interpret how increased rest or more intense exercise contributes to changes in memory performance, controlling for age.

4.2 Sample Size Determination

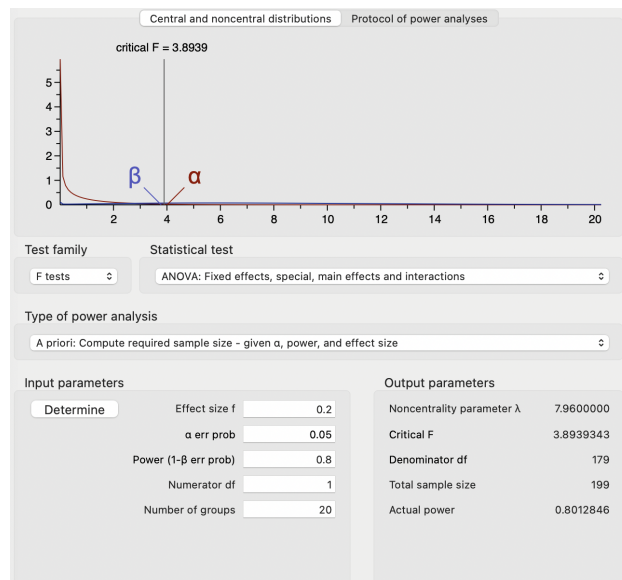


Figure 3: G*Power Output

To determine an appropriate sample size, we used G*Power with a power level of 0.80, a significance level of 0.05, and an effect size of 0.2. Given our 2x2 factorial design and the inclusion of 5 age-based blocks, we needed to ensure that each treatment combination was equally represented across all age groups. This resulted in four treatment combinations and five age blocks, totaling 20 unique groups. To maintain balance and sufficient power for detecting interaction and main effects, we assigned 10 participants to each group, leading to a total sample size of 200 participants. This sample size provides adequate sensitivity to detect meaningful differences in memory performance while maintaining the structure of our randomized complete block design.

5 Results

5.1 Box Plots

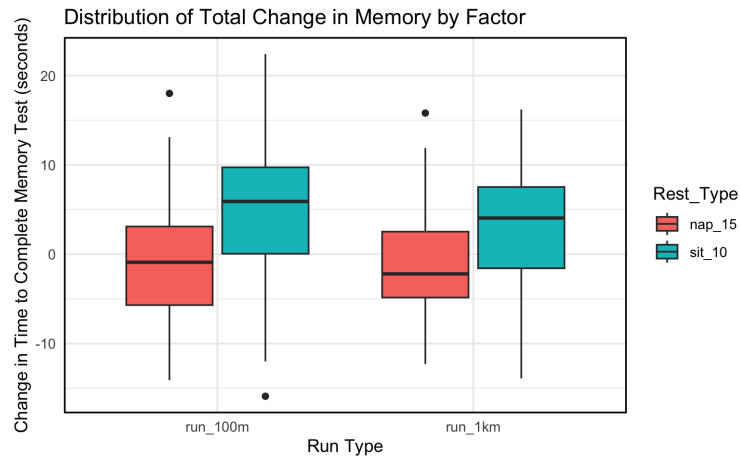


Figure 4: Box Plot of Change in Memory by Factor

The distributions of each group show that only the two nap groups are centered around negative values, which may imply only the nap condition has a positive effect on memory. The sit condition may even worsen memory performance since the distributions show a positive difference from memory performance before and after experimental intervention. There looks to be no significant main effect of the run distance since the distributions look similar across run types.

5.2 T-Test

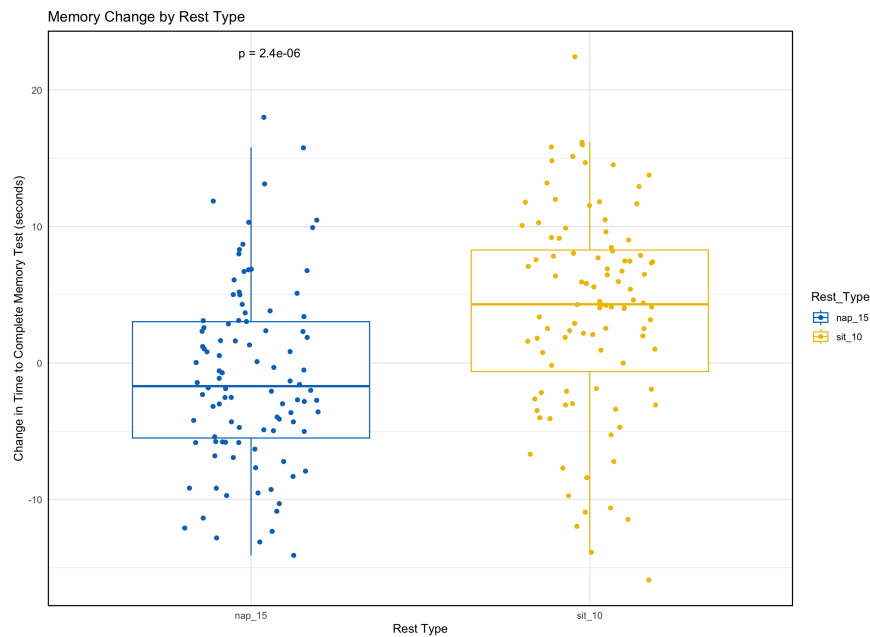


Figure 5: T-Test: Rest Type

From the plot, the nap 15 group has a lesser mean than the sit 10 group. This indicates that participants who napped for 15 minutes improved their memory more compared to those who just sat to rest.

The confidence intervals for the two groups are well-separated, and the t-test confirms that this difference is statistically significant. Therefore, we conclude that napping led to significantly greater improvements in memory performance (as measured by time reduction) than sitting as a form of rest.

5.3 Interaction Plots

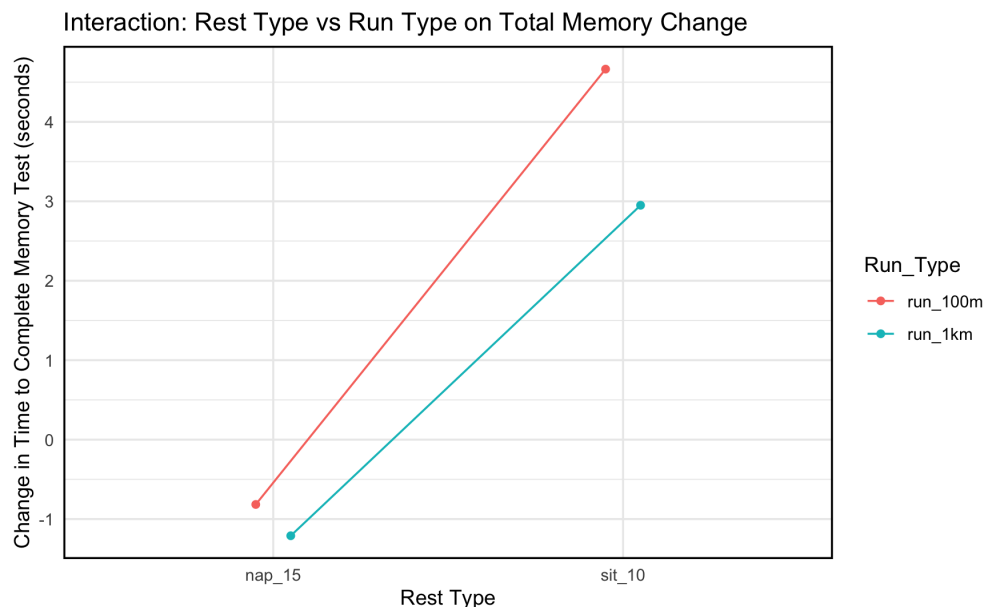


Figure 6: Interaction Plot: Rest Type vs. Run Type

The interaction plot between Rest Type and Run Type suggests there is likely no interaction effect between these two factors. This implies that how someone rests and the intensity of subsequent exercise together don't impact memory retention. We will explore this interaction in more detail in the ANOVA table below.

5.4 ANOVA Analysis

Table 1: ANOVA Summary Table

Source	DF	Sum Sq	Mean Sq	F-Value	Pr(>F)
Rest_Type	1	1162	1161.6	24.327	1.76e-06
Run_Type	1	56	55.5	1.163	0.2821
Age_Group	4	502	125.4	2.627	0.0359
Rest_Type:Run_Type	1	22	21.8	0.456	0.5003
Residuals	192	9168	47.8		

Based on the ANOVA table above, we see that the p-value for both the Run Type ($p = 0.2821$) and the p-value for the interaction between the Rest Type and the Run Type ($p = 0.5003$). This confirms what we explored in the interaction plot above; there is no significant interaction between Rest Type and Run Type. It goes further to show us that Run Type alone is also not significant in affecting the speed at which a participant is able to complete a memory test. However, the ANOVA table also shows that the age group has a significant p-value ($p = 0.0359$), indicating that it is a successful blocking factor. It also shows that Rest Type has a significant p-value ($p = 1.76e-06$), indicating that Rest Type does make a statistically significant impact on the change in the speed at which a participant is able to complete a memory test.

5.5 Tukey Post-Hoc Analysis

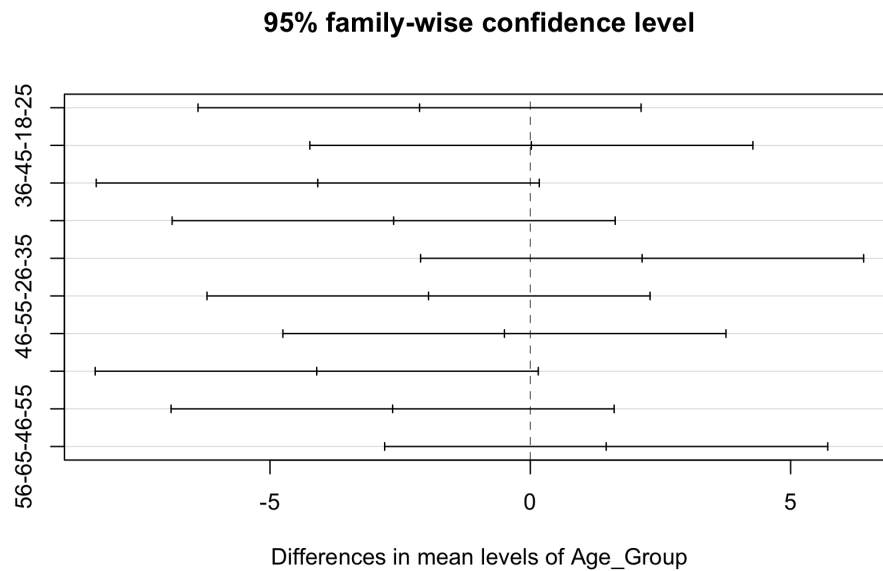


Figure 7: Tukey HSD: Age Group

Looking at the Tukey Plot above, showing the distributions of the change in time (in seconds) of completing the memory test before vs after the treatments between age groups. We can see that all of the intervals include zero. This indicates that there are no statistically significant pairwise differences between individual age groups, even though the overall ANOVA showed that Age Group has a significant effect.

5.6 Residual Diagnostics

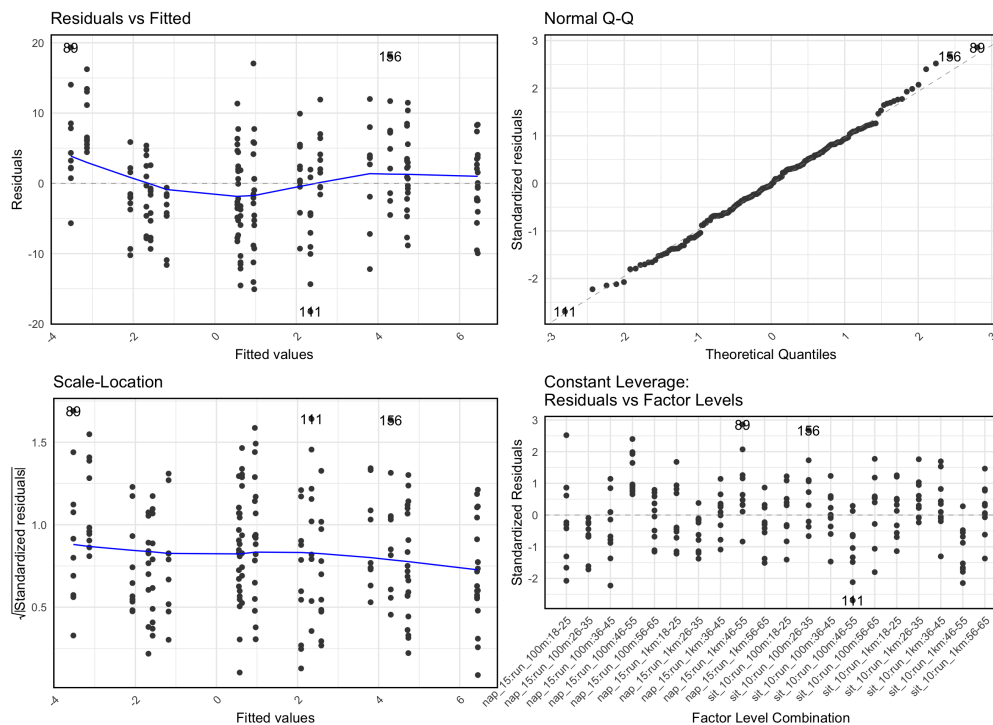


Figure 8: Residual Diagnostics

Each of these four plots including Residuals vs Fitted, Normal Q-Q, Scale-Location, and Constant leverage further suggests that the residuals of the proposed model remain constant. This validation suggests that the assumption of constant variance is maintained and the model is valid indicating significant and reliable findings.

6 Discussion

The purpose of this study was to evaluate how short-term rest and physical activity influence memory performance, and whether their effects vary across different age groups. Our findings provide strong support for the hypothesis that rest type—specifically, taking a brief nap—has a statistically significant effect on memory, as measured by completion time on a card-matching task. Rest type had a highly significant effect ($F = 24.3$, $p < 0.001$), whereas run type showed no significance ($F = 1.16$, $p = 0.28$). Participants who napped for 15 minutes outperformed those who sat quietly for 10 minutes, suggesting that even a short period of sleep may facilitate the consolidation of recently encoded information. In contrast, run type did not have a statistically significant effect on memory performance, nor was there a significant interaction between rest and run types. This suggests that while physical activity may provide long-term cognitive benefits, a single bout of brief exercise may not be sufficient to produce measurable improvements in memory in a short-term testing environment. The lack of a significant interaction further indicates that the cognitive benefits of rest are not enhanced or diminished by the intensity of exercise performed prior to testing. One limitation of this study is that memory was measured using a single task type, which may not generalize to other cognitive domains such as recall or spatial memory. Additionally, short-

term interventions like a 15-minute nap or 1km run may not capture the full potential of long-term lifestyle impacts on memory.

Notably, the inclusion of age group as a blocking factor was validated by our analysis. The statistically significant p-value for age demonstrates that cognitive performance differed across age groups, and that blocking effectively controlled for this variability. This reinforces prior research indicating that memory function changes with age, and underscores the importance of accounting for age-related differences in cognitive studies. Residual diagnostic plots supported the validity of the ANOVA model. The assumptions of homogeneity of variance and normality appeared to hold, strengthening the reliability of our statistical conclusions. Furthermore, the box plots and interaction plots provided visual confirmation of our findings—particularly the consistently superior performance of the nap condition and the minimal differences between the two run conditions. The boxplots showed that the nap group consistently had lower median times across age groups, while the sit group displayed greater variability, suggesting more inconsistent memory performance. Using G*Power with a power level of 0.80, a significance level of 0.05, and a conservative effect size of 0.2, we calculated a required sample size of 199 to detect treatment effects with high confidence. We easily rounded this to 200, above the minimum sampling size but very close, allowing for a balanced 2x2 factorial design across five age blocks and ensuring sufficient replication to support valid inference across all conditions.

Together, these results suggest that rest, rather than exercise, has the more immediate and pronounced effect on memory performance under the conditions tested. While physical activity remains an important contributor to cognitive health over the lifespan, our findings support the role of short naps as a valuable and easily implementable tool for enhancing memory in the short term. Future research should explore how longer durations of rest or more sustained exercise interventions influence memory over time, and whether combining these interventions sequentially (e.g., exercise followed by a nap) might lead to synergistic cognitive benefits. It may also be useful to investigate other cognitive domains - such as attention or problem-solving - under similar experimental conditions. Furthermore, it could explore whether a sequence of exercise followed by rest yields greater benefits, or examine the long-term cognitive effects of consistent napping or physical activity. Lastly, studies should consider extending the age range to include children or older adults to more fully characterize developmental and aging-related patterns in cognitive responsiveness to lifestyle interventions.

7 References

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